

3(a)(i)	amplitude = 0.020 m	A1
3(a)(ii)	$T = 0.60 \text{ s}$	C1
	$f = 1 / T$ $= 1.7 \text{ Hz}$	A1
3(a)(iii)	$a = (-)\omega^2 x$ and [$\omega = 2\pi f$ or $\omega = 2\pi / T$]	C1
	$a = (4\pi^2 / 0.60^2) \times 2.0 \times 10^{-2}$ $= 2.2 \text{ m s}^{-2}$	A1
3(b)	$1.67 = (1 / 2\pi) \times [(24 \times 10^{-4} \times \rho \times 9.81) / 0.23]^{1/2}$	C1
	$\rho = 1.1 \times 10^3 \text{ kg m}^{-3}$	A1
3(c)	wave starting with a peak at (0,6)	B1
	wave with same period (or slightly greater)	B1
	peak height decreasing successively	B1